What the Kostant–Michor citation trail actually tells us

Cayley maps, related theorems, and a focused research program

Research report prepared for Sean Rogers

27 October 2026

Abstract

This report examines twelve identified citing works, the original Kostant–Michor paper, and two later papers needed to correct the graph-theoretic branch of the citation trail. It distinguishes trace-projection Springer morphisms, classical matrix Cayley transforms, arbitrary equivariant rational maps, and local coordinates. The strongest connections concern invariant gradients, harmonic localization, and the actual fibers of prescribed Springer maps. Our recent results are described at their repository proof status, with remaining review obligations explicit. The recommended next theorem is a structural classical classification of generic nonemptiness and excess under polynomial operations on the adjoint character.

1 Conclusions and scope

The citation list contains several different mathematical stories. It should not be treated as a list of open problems awaiting our Springer-map calculations.

1. **The closest connections are substantive.** Lemire–Popov–Reichstein study equivariant birational geometry and degree; Agricola–Goodman study invariant vector fields and harmonic separation; Lu–Ye classify representations admitting the classical matrix transform. They provide useful tools and boundaries on novelty claims.
2. **Our strongest current extension concerns a fixed map and actual fibers.** The repository contains a case proof of generic smoothness of regular adjoint Springer fibers over centralizer-exact Levi strata, followed by nonemptiness and representation-composition results. These differ from global regularity preservation, birationality, and integral Borel images.
3. **The harmonic connection is concrete.** For the symplectic determinant-character family, separation works after a smaller localization than the full Jacobian localization. Sufficiency is largely a consequence of known separation; sharpness is the additional argument to review.
4. **Other citations have weaker implications.** The physics papers use optional local coordinates or a metaplectic formula. The graph paper uses an exponential inequality; later papers correct its broad claims. The torsion survey has a bibliographic citation without a located map-dependent argument.

Coverage. The twelve profiles cover the eleven previously identified works, including the dissertation, plus Lu–Ye, an additional direct citer found in this audit. Preprint and journal versions count once. Citation indexes were used for discovery and accessible full texts for verification. Index coverage is incomplete and sometimes duplicates versions: this is not a claim to have found every citation worldwide. The relevant arguments and surrounding hypotheses were examined, not every proof in the long physics and differential-geometry papers.

Evidence labels. "Published" means located in the cited work. "Repository deduction" means a written argument in our project, not external approval or an established priority claim. "Elementary comparison" means an argument supplied here. "Proposed" means unfinished work. The manifest records source versions and local proof-note hashes. Nothing below establishes a new minimum Cayley degree or an application resolving a physics problem.

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2 Four meanings of Cayley map

Let G be a connected complex semisimple group, $T \subset G$ a maximal torus, and W its Weyl group. Let $\rho : G \to \mathrm{GL}(V)$ have injective differential. Put

$$B_\rho(X, Y) = \mathrm{tr}(\mathrm{d}\rho(X)\mathrm{d}\rho(Y)).$$

The normalized Springer–Cayley morphism is characterized by

$$B_\rho(\theta_\rho(g), X) = \mathrm{tr}(\rho(g)\mathrm{d}\rho(X)). \tag{1}$$

It is regular and conjugation-equivariant, with $\theta_\rho(e) = 0$ and $\mathrm{d}\theta_{\rho,e} = 1$. For simple G, fix an invariant form B and write $B_\rho = I_\rho B$. This scalar must be retained when comparing representations. For the adjoint map take B to be the Killing form, so $I_{\mathrm{Ad}} = 1$.

Trace projection.
Equation (1), defined everywhere and generally of degree greater than one.

Classical matrix transform.
On suitable matrix groups,

$$C(g) = (g - I)(g + I)^{-1}, \qquad C^{-1}(X) = (I + X)(I - X)^{-1}.$$

These are rational inverses on specified opens. Applicability depends on the representation.

Equivariant rational Cayley map.
An arbitrary dominant G-equivariant rational map $G \dashrightarrow \mathfrak{g}$. Its smallest possible degree is $\mathrm{Cay}(G)$. It need not come from a character or be regular everywhere.

Local coordinate map.
An analytic or formal map $\mathfrak{g} \to G$ with invertible differential. The exponential and a local inverse to (1) are examples. A formal inverse does not imply a rational inverse.

An element g is regular when $\dim Z_G(g) = \mathrm{rank}\, G$. The analogous condition defines regular elements of $\mathfrak{g}$. A regular morphism need not preserve regular elements.

For a parabolic P with Levi L, the Kumar construction is

$$\underbrace{\mathrm{Rep}^{\mathbb{C}}_{\rho\text{-poly}}(L)}_{\theta^*_{\rho,T}\mathbb{C}[\mathfrak{t}]^{W_L}} \xrightarrow{(\theta^*_{\rho,T})^{-1}} \mathbb{C}[\mathfrak{t}]^{W_L} \xrightarrow{\beta^P} H^*(G/P, \mathbb{C}). \tag{2}$$

The first arrow is the inverse of an injective algebra map *on its image*, not a geometric inverse to $\theta_{\rho,T}$. Integral versions require a specified lattice and normalization before applying β^P. Smooth group fibers do not determine Smith factors of a Borel image.

2.1 The Kostant–Michor baseline

Source and location: [1], ğğ2–5, especially Theorem 3.2, Corollary 3.3, and ğğ4.11–4.13. Kostant–Michor develop trace projection, its differential, quotient degree, orbit behavior, and harmonic separation. For the harmonic space $\mathcal{H}$ on $\mathfrak{g}$ and $\Psi = \det \mathrm{d}\theta$, their construction gives separation using $\theta^*\mathcal{H}$ after Jacobian localization. The corresponding global multiplication isomorphism is tied to preservation of regular elements. They establish the standard SL_n case. Their treatment of hyperbolic elements includes the real exponential estimate later improved by Borwein–Girgensohn.

Our extension questions concern smaller opens supporting this *fixed pulled-back* harmonic space, and actual regular fibers over prescribed nonregular target strata. Existence of some harmonic complement does not settle either question. Conversely, generic smoothness over a Levi center does not give global regularity preservation or global harmonic multiplication.

3 Paper-by-paper analysis: the algebraic connections

3.1 Lemire–Popov–Reichstein: *Cayley groups*

Source and location: [2], arXiv v2, Theorems 1.28 and 1.31; ğ10, especially Example 10.4, Proposition 10.7, and Definition 10.9. Journal numbering can differ.

Actual use. This paper relates equivariant birationality to Weyl-group character lattices and classifies simple Cayley groups. Trace projection is a source of dominant equivariant morphisms, but the permitted rational maps form a much larger class. The paper also discusses clearing invariant denominators and already defines Cayley degree; that definition should not be credited exclusively to the later degree paper.

What this means for us. Our torus formulas give a controlled family inside their broader problem. Complete classification of our family's degrees would not classify all equivariant rational maps. Likewise, the root and weight lattices in our Borel work and those in stable birational classification have a common origin but encode different obstructions. A Smith factor in cohomology is not automatically a birational obstruction.

The distinction between Spin_n and SO_n matters. A finite central quotient preserves certain local geometric properties in characteristic zero but changes the character lattice and can change degree. A statement about a matrix image cannot silently become a birational statement about its simply connected cover.

Possible extension. Characterize which low-degree Weyl-equivariant rational maps can arise, after invariant scalar modification, from a character gradient. Equivariance is insufficient: the corresponding invariant one-form must be exact, and actual representations impose nonnegative integral multiplicities. Our composition theorem treats a special exact family, not all denominator-clearing operations.

Assessment. High conceptual relevance, but a difficult realization problem. Use the paper to make degree and lattice claims precise rather than promise to solve its classification questions from our fiber theorem.

3.2 Lemire–Popov–Reichstein: *On the Cayley degree of an algebraic group*

Source and location: [3], Theorems 2–3; ğ3, Lemma 2 and Corollary 2; ğğ4–6.

Actual use. They prove $\mathrm{Cay}(\mathrm{SL}_n) \le n-2$ for $n \ge 3$ and $\mathrm{Cay}(G_2) = 2$. Their torus reduction identifies the degree of an equivariant map with its Weyl-equivariant torus-map degree. The G_2 construction uses rational torus coordinates, a quadric, and projection; it does not assert that a preferred trace-projection representation has degree two. Their final example also treats a higher-degree torus formula.

Our comparison. For any specified representation,

$$\deg \theta_\rho = [\mathbb{C}(G) : \theta_\rho^* \mathbb{C}(\mathfrak{g})], \qquad \mathrm{Cay}(G) \le \deg \theta_\rho.$$

The first number measures our map; the second minimizes over all admissible rational maps. Boundary corrections in our exceptional computations determine the first number. They improve an upper bound

for the second only if they beat an existing bound. They give no lower bound without excluding *every* smaller-degree rational map.

Our degree records include 1080 and 2754 for the simply connected E_6 minimal and adjoint maps, and 7272 and 33664 for their E_7 counterparts. These are repository calculations with specified lattice-cover factors and boundary corrections. They are not values of $\mathrm{Cay}(E_6)$ or $\mathrm{Cay}(E_7)$, or assertions about all fibers. This review does not rerun those proofs.

Proposed extension. A deformation theorem could explain when generic degree is constant in a character-gradient family and how contributions at infinity change it. It would require a specified compactification, scheme multiplicities at infinity, and central-isogeny factors. This is closer to our tools than universal Cayley-degree lower bounds. It is separate from the recommended next classical nonemptiness theorem. Historical minimum-degree questions should not be labelled currently open without a further status search.

3.3 Agricola–Goodman: invariant vector fields on symmetric spaces

Source and location: [4], author version of 28 March 2003, ğ4.1, Theorem 4.1; ğ4.2, Theorems 4.2–4.3; ğ5.2, Theorem 5.3.

Actual use. For semisimple simply connected G, gradients of fundamental characters freely generate conjugation-invariant vector fields over $\mathbb{C}[G]^G$. For SL_n explicit generators are

$$\Phi_k(g) = g^k - \frac{\mathrm{tr}(g^k)}{n} I, \qquad 1 \le k \le n-1.$$

The paper gives commutator formulas and an invariant differential operator yielding harmonic separation for SL_2. Its symmetric-space "flat" vector fields concern a lifting condition, not flatness of our fiber families.

Exact bridge. Write $\chi_\rho = F(\chi_1, \ldots, \chi_r)$ in fundamental characters. With the trace-form normalization,

$$I_\rho \theta_\rho = \sum_i I_i \frac{\partial F}{\partial \chi_i} \theta_i. \tag{3}$$

This is the chain rule applied to known invariant-gradient theory. Implementing it is useful, but is not a new basis theorem. An arbitrary invariant linear combination of gradients need not itself be a character gradient: its coefficient one-form must satisfy integrability.

Where our work goes further, subject to review. We ask how these coefficients control the actual target-fiber scheme. For $F = f(\chi_{\mathrm{Ad}})$, our representation-composition theorem turns the chain rule into a criterion for retaining or erasing dominating critical components. The chain rule is published background; the component-survival deductions are ours. Persistent nilpotents make reduced-support computations insufficient.

Proposed extension. Classify constant adjoint-character values on central critical components from classical block data. This is a structural question in unbounded rank. It uses the known gradient module to explain geometry, rather than re-proving that module's existence. A later operator question is whether the fixed harmonic decomposition admits annihilating operators without unnecessary Jacobian poles. Multiplication alone does not remove inverse-differential poles.

3.4 Lu–Ye: *The Cayley Transform on Representations*

Source and location: [5], v2, 18 July 2025; Definitions 4.1 and 4.5, Theorem 4.9, ğ5, and Theorems 6.13 and 6.15.

Actual use. They ask when the classical transform applied to $\mathrm{d}\rho(\mathfrak{g})$ lands in $\rho(G)$. They give power-closure and weight-geometric criteria, then representation classifications in the complex simple and compact settings. Familiar orthogonal and symplectic images, low-rank identifications, and type-D_4 half-spin representations occur. Kostant–Michor is cited, including in the unipotent-group discussion.

A genuinely different property. Their applicability condition concerns a rational matrix formula. Our regularity question concerns the global trace-projection morphism. Neither property implies the other, as the following elementary comparisons show.

For defining SL_3,

$$\theta(g) = g - \tfrac{1}{3}\operatorname{tr}(g)I.$$

Subtracting a scalar leaves the matrix centralizer unchanged, so regularity is preserved. But for $X = t\,\mathrm{diag}(1, 1, -2)$,

$$\det((I + X)(I - X)^{-1}) = \frac{(1 + t)^2(1 - 2t)}{(1 - t)^2(1 + 2t)}$$

is not identically one. The classical formula does not generically land in SL_3.

Conversely, defining Sp_4 admits the classical transform, but its trace projection is $(g - g^{-1})/2$. The torus element

$$g = \mathrm{diag}(2, -1/2, 1/2, -2)$$

is regular semisimple, whereas its image

$$\theta(g) = \mathrm{diag}(3/4, 3/4, -3/4, -3/4)$$

is nonregular in $\mathfrak{sp}_4$. The accompanying script checks this using exact matrices and root values.

Research implication. A regularity-classification paper must explicitly compare these two properties and distinguish G from its matrix image when the representation has a central kernel. A joint description of classical applicability, trace-projection regularity, and harmonic localization would be meaningful. Another isolated representation calculation would not provide it.

4 The inequality branch

4.1 Borwein–Girgensohn: *A class of exponential inequalities*

Source and location: [6], introduction and theorem of ğ2; extensions in ğğ3–4.

Actual use. This paper improves the real estimate in Kostant–Michor. For real x_i with $\sum_i x_i \geq 0$ it proves

$$\sum_{i=1}^{N} x_i e^{x_i} \geq \frac{C_N}{N} \sum_i x_i^2, \qquad C_N = \max\{2, e(1 - 1/N)\}. \tag{4}$$

There are other norm and integral versions. The paper does not classify complex fibers or representations.

Immediate application. Let H lie in a real Cartan where the weights μ_i of ρ, repeated with multiplicity, are real. Semisimplicity gives $\sum_i \mu_i(H) = 0$. Then

$$B_\rho(\theta_\rho(e^H), H) = \sum_i \mu_i(H) e^{\mu_i(H)}, \quad B_\rho(H, H) = \sum_i \mu_i(H)^2.$$

Substitution in (4) gives a quantitative coercive estimate. This is a direct application of the published inequality, not a new Springer-map theorem.

For a self-dual weight multiset, pairing $a = \mu(H)$ with $-a$ and using $a \sinh a \geq a^2$ gives the elementary stronger bound

$$B_\rho(\theta_\rho(e^H), H) \geq B_\rho(H, H).$$

The Hessian of $H \mapsto \chi_\rho(e^H)$ is likewise at least B_ρ, since each paired contribution has $\cosh \mu(H) \geq 1$. This supplies real strong monotonicity, not a new classification.

Limit. Complex weights and exponentials do not retain this positivity. Cancellations can create singular or nonreduced fibers. These estimates therefore do not supply the missing uniform exceptional identity. A real numerical-continuation application is plausible, but is a separate objective with additional quantitative work.

4.2 Estrada–de la Peña: maximum walk entropy

Source and location: [7], Theorems 2.1–2.2 and ğ4, especially Remark 4.5(b); follow-ups [8, 9].

Actual use. This studies entropy of the normalized diagonal of $e^{\beta A}$ for an adjacency matrix A. The Kostant–Michor connection is an exponential inequality, sharpened by Borwein–Girgensohn. It does not construct a group-to-algebra map. The Cayley–Hamilton theorem mentioned nearby is a different result.

Published correction. Kloster–Král'–Sullivan exhibit a non-walk-regular, non-degree-regular graph on 24 vertices with maximal walk entropy at some $\beta > 0$, and identify a flaw in the earlier proof. Ballini–Deniskin prove the characterization for every nonzero algebraic β, including $\beta = 1$, using Lindemann–Weierstrass. Thus the unrestricted-temperature claim fails; one must not say that the temperature-one statement is false.

Independent warning. A uniform positive entropy gap as $\beta \downarrow 0$ is impossible for a fixed nonregular graph: $e^{\beta A} \to I$, normalized diagonal entries tend to $1/n$, and entropy tends to $\log n$. This checks the overstrong formulation of Theorem 2.2(c) without using the later counterexample.

Ramification. There is no established route from our complex fiber theorem to a new graph-entropy theorem. The inequality is the actual connection, and the graph question has its own corrected literature. Record it accurately and leave it outside the main agenda.

5 Local coordinates, quantization, and physics

5.1 Cariñena et al.: *Hopf algebras in dynamical systems theory*

Source and location: [10], Appendix C, equations (C.2)–(C.3), PDF pages 52–53 of v2.

Actual use. In the discussion of differential equations on Lie groups, a local map $f : \mathfrak{g} \to G$ may be exponential or a suitable Cayley parametrization. Pullback converts a group-valued equation to one on the algebra, with the Darboux derivative controlling the conversion. This is a local-coordinate use, not global fiber classification.

Our available contribution. Identity differential at e gives a unique formal inverse to θ_ρ, and an analytic local inverse. Equivariance follows from uniqueness. Our weight and root-commutator formulas can produce such coordinates without convenient representation matrices. Existence is the inverse-function theorem, not new mathematics.

For $F_\rho(X) = \theta_\rho(e^X)$, direct expansion gives

$$B_\rho(F_\rho(X), Y) = B_\rho(X, Y) + \tfrac{1}{2} \operatorname{tr}(\mathrm{d}\rho(X)^2 \mathrm{d}\rho(Y)) + O(X^3).$$

If the quadratic term is $A_2(X)$, the inverse begins $X = Z - A_2(Z) + O(Z^3)$. Our Borel-coordinate recurrence can organize higher terms. A useful application would require error or convergence bounds and a demonstrated integrator benefit. Smooth positive-dimensional fibers away from the identity do not define coordinate charts.

Assessment. A plausible later application; lower priority than a structural geometric theorem.

5.2 Gracia-Bondía et al.: *Infinitely many star products to play with*

Source and location: [11], Appendix A, equations (A.12)–(A.14), PDF pages 32–34 of v2.

Actual use. A metaplectic-symbol formula uses $(I - S)(I + S)^{-1}$ in a Gaussian and $\det(I + S)^{-1/2}$ in its amplitude, with a sign or phase from the metaplectic cover. The boundary $\det(I + S) = 0$ needs distributional treatment. This is a concrete classical Cayley application.

Overlap. The same divisor $D(g) = \det(I + g)$ occurs in our symplectic harmonic-localization theorem. Both constructions simplify in classical Cayley coordinates. However, the metaplectic caustic and the full critical locus of our polynomial map need not coincide: the latter has an additional factor even on $D \neq 0$.

Possible extension and cost. A character-dependent scalar rescaling might suggest alternative symbols. Substitution into the Gaussian is not a proof: composition under the star product, phase, normalization, and distributional extension all need new analysis. Our localization result supplies none of these automatically. This is a potential collaboration project with a real shared boundary, not an immediate quantization corollary.

5.3 Borsten et al.: *Double Copy from Homotopy Algebras*

Source and location: [12], ğ3.2, PDF page 19, footnote 18.

Actual use. The nonlinear sigma-model discussion uses exponential coordinates and mentions the Cayley alternative. Its treatment is independent of the choice under the stated equivariance condition. The homotopy-algebra construction does not depend on classifying global Cayley fibers.

Comparison. Our formal inverse supplies another local parametrization. That is a coordinate option. Global algebraic multiplicity and remote singular fibers are invisible to a finite perturbative expansion at the identity. A meaningful application would identify an improved quantity, such as interaction-vertex complexity, and justify the field redefinition.

Assessment. No current result of ours resolves an open double-copy assertion located here. The useful outcome is a precise boundary on the connection, not a claim about scattering amplitudes.

5.4 Borsten et al.: tree-level to loop-level color–kinematics duality

Source and location: [13], ğ4.1, PDF pages 19–20, footnote 16.

Actual use. Cayley coordinates are again an alternative to exponential coordinates for a group-valued sigma-model field. The main result concerns passing from tree-level duality to loop-level duality up to counterterms. The coordinate choice is ancillary.

Ramification. Our formulas might reorganize vertices, but smoothness of a complex group fiber does not remove quantum counterterms, prove loop equivalence, or justify a measure-level field redefinition. Those are separate arguments. Even a simple rational inverse would not automatically settle them.

Assessment. Keep the exact citation, but no new project is justified here by our present theorem alone.

6 The survey citation and your original work

6.1 Agricola: the Srní lectures on geometries with torsion

Source and location: [14], journal PDF and arXiv version; bibliography entry [KM01].

What was found. Both versions include Kostant–Michor in the bibliography. The audit did not locate a body argument using its map. The survey concerns torsion, spin geometry, and Dirac operators. "Cayley plane" is not a Cayley transform.

Interpretation. This is a verified bibliographic citation without an identified substantive map application. We cannot infer that our fibers extend a torsion or Dirac theorem because spin representations or Kostant's name occur in both subjects. The cubic Dirac operator is another line of work.

Assessment. A project here first needs a specific map-dependent assertion. The citation alone is not evidence for one.

6.2 Rogers: dissertation, explicit formulas, and Springer–Borel maps

Source and location: [15], Chapter 3, printed pages 26–30; [16]; cohomology applications in [17, 18, 19].

Actual use. The dissertation imports basic Springer-map properties, the differential, and quotient degree from Kostant–Michor, then develops torus formulas and representation-ring applications. The explicit-determination paper gives the weight sum in fundamental-weight/simple-coroot conventions. The Kumar series uses pullback to define the polynomial representation source and then applies the Borel homomorphism.

The durable computational bridge. Write each weight as the column of its values on the simple coroots, with multiplicity m_μ, and set

$$M_\rho = \sum_\mu m_\mu \mu\mu^{\mathsf{T}}, \qquad v_\rho(t) = \sum_\mu m_\mu \mu e^\mu(t).$$

For $\theta_\rho(t) = \sum_i c_i(t)\alpha_i^\vee$,

$$M_\rho c(t) = v_\rho(t). \tag{5}$$

Infinitesimal faithfulness makes M_ρ invertible. Thus one computes the map from characters without full representation matrices. Differentiation gives the torus differential; equivariance controls root directions, with extension across denominator loci requiring justification.

Our extension. We use this same map to study fibers, critical images, and representation dependence, and separately to study integral sources and Borel images. These are two branches of one construction, not one theorem. Function-field degree does not determine a Smith form; a complex representation-ring surjection does not establish fiber smoothness.

Source corrections. Two Chapter 3 passages require correction: a Jacobian notation error and an overstatement about Jordan decomposition. Appendix A gives exact checks. The historical dissertation is unchanged. This is not evidence that its downstream results all fail. The accessible preprint of [16], rather than an inaccessible final journal PDF, was used for that paper's comparison.

7 What our current results actually add

7.1 Regularity preservation: a separate classification

Status: repository uniform argument with internal checks; not externally reviewed or established novel.

For simple simply connected complex groups and actual nontrivial representations, allowing direct sums and trivial summands, the current classification note asserts that global regularity preservation occurs only for copies of the defining representation of SL_N, or copies of its dual, with optional trivial summands. For $N \geq 3$ a mixture of defining and dual nontrivial summands is excluded. The low-rank identifications are understood.

Torus regularity is weaker. In the symplectic family it also permits actual characters $A\det(I+g)^m + B$ with positive integers A, m and an integer B making the character actual. For rank at least two these do not preserve global regularity. The remaining simple types have no positive entries in the note's table. Virtual characters, positive characteristic, and general reductive products are not classified.

This is the classification to compare with Lu–Ye, not the adjoint fiber theorem below. Its proof uses positive weight multiplicities and root-hypertorus factorization; it does not follow just from a list of irreducibles. Its novelty and completeness require focused review, and this report has not independently re-proved that table.

7.2 The adjoint generic regular-fiber theorem

Status: repository case proof and internal audits; external mathematical review remains necessary.

Let G be simple and simply connected, and $\theta = \theta_{\mathrm{Ad}}$ use the Killing form. For a proper Levi K, put

$$Z_K^\circ = \{h \in \mathfrak{z}(\mathfrak{k}) : Z_G(h) = K\}.$$

The consolidated repository theorem asserts a dense open $U_K \subset Z_K^\circ$ such that

$$\theta^{-1}(h) \cap G_{\mathrm{reg}} \quad \text{is smooth as its actual scheme for every } h \in U_K. \tag{6}$$

The case proof covers A–D, G_2, F_4, E_6, E_7, E_8. Empty fibers and positive-dimensional transverse factors are permitted; regular nonsemisimple source points remain included. A separate note extends the adjoint assertion to semisimple groups and full-factor zero targets. That extension cannot automatically be transferred to other characters.

This differs from ordinary generic smoothness over the whole target: a Levi center is a special lower-dimensional locus that singular or nonreduced components could dominate. It also differs from regularity preservation, since h is generally nonregular in $\mathfrak{g}$. Smoothness in (6) does not imply invertibility of $\mathrm{d}\theta$ along the fiber.

Common mechanism. On a finite central cover of K, choose coordinates $x_1, \ldots, x_\ell$ given by fundamental characters on the derived quotient, and central torus coordinates $z_1, \ldots, z_s$. Write the full restricted adjoint character as $\Phi(x, z)$ and $D_j = z_j \partial_{z_j}$. Then

$$Y = V(\Phi_{x_1}, \ldots, \Phi_{x_\ell}), \qquad \mu = (D_1\Phi, \ldots, D_s\Phi).$$

The central fiber equations are $\Phi_{x_i} = 0$ and $D_j\Phi = c_j$, where c is an invertible linear reparametrization of h. At the relevant regular points the completed group-fiber ring is the completed critical-fiber ring with $\dim K - \operatorname{rank} G$ formal smooth variables adjoined. Equality of ideals and descent through the central cover are essential; agreement of point sets would lose nilpotents.

Review boundary. The final E_8 proof includes exact algebraic certificates. A fresh audit independently rebuilt characteristic-zero identities and portions of the degree argument. Some finite-prime ideal memberships still rely on one pinned F4 engine, rather than independent stored multiplier certificates. This is a specific review obligation. A certificate is replayable evidence for specified algebraic claims, not a substitute for checking their geometric implications.

7.3 Nonemptiness, excess, and representation composition

Status: repository deductions GEO-93–96; the ordinary adjoint classification was already complete at GEO-49.

For a generically actual-smooth critical component Y_i, our critical-value theorem identifies

$$\overline{\mu(Y_i)} = (\operatorname{Lie} H_i)^{\perp}.$$

Here H_i is the connected horizontal stabilizer of the critical-value variety recording (z, Φ): translations in the central torus that preserve that variety and fix its scalar-value coordinate. Dominance is equivalent to trivial connected stabilizer. On a dominant component, transverse excess is $\dim Y_i - s$. This is a criterion, not a root-theoretic classification of the components or stabilizers. Its actual-smoothness hypothesis cannot be dropped.

For an actual representation σ with $\chi_\sigma = f(\chi_\rho)$ and $f'(\dim \rho) \neq 0$, normalization and the chain rule give

$$\theta_\sigma = \frac{f'(\chi_\rho)}{f'(\dim \rho)} \theta_\rho. \tag{7}$$

The formula itself is published-background calculus. The further proper-Levi result for $\rho = \mathrm{Ad}$ says that a dominating component survives generically unless χ_{Ad} is constant there with value b and $f'(b) = 0$. Nonconstant components are not erased in this way. Thus changing representation can change nonemptiness despite equivariance and identical local normalization at the identity.

An unbounded-rank family is already available: the GL_4^q Levi pattern in B_{4q}, C_{4q}, D_{4q} has constant values

$$-\frac{q}{2q-1}, \qquad 0, \qquad 0,$$

respectively, and adjoint generic regular fibers of dimension $12q$. This denotes the matrix block pattern; the finite central cover remains relevant for simply connected groups. Pure adjoint tensor powers of order at least two erase these nonzero-target strata in C, D; a positive shifted-adjoint construction does so in B. This is not the classification for arbitrary block partitions.

The E_6 contact example is a separate control: its smooth generic group fiber has dimension 32 and transverse excess two. A shifted adjoint tensor construction can erase it. Thus universal quasifiniteness cannot replace our target theorem.

7.4 Sharp harmonic localization

Status: repository deduction; sufficiency follows formally from known separation, while sharpness and the Jacobian have separate arguments.

Let $G = \mathrm{Sp}_{2n}$, $n \geq 2$, $D(g) = \det(I + g)$, and $m \geq 1$. The determinant-character family has normalized map

$$\theta_m(g) = \frac{2}{4^{nm}} D(g)^{m-1} (g - I) \operatorname{adj}(g + I).$$

For the graded harmonic vector space $\mathcal{H} \subset \mathbb{C}[\mathfrak{g}]$, set $\mathcal{H}_m = \theta_m^* \mathcal{H}$. The repository proves multiplication gives

$$\mathbb{C}[G]^G[D^{-1}] \otimes_{\mathbb{C}} \mathcal{H}_m \xrightarrow{\sim} \mathbb{C}[G][D^{-1}], \tag{8}$$

and asserts the analogous localization at a nonzero invariant f works exactly when $D \mid f$. This is a module and G-representation decomposition; the harmonic space is not being called an algebra.

The sufficiency mechanism is simple. In classical Cayley coordinates $X = (g - I)(g + I)^{-1}$, let $\delta = \det(I - X)$. Then

$$D = 4^n/\delta, \qquad \theta_m = 2\delta^{-m} X.$$

A homogeneous harmonic of degree d is rescaled by the invariant unit $2^d \delta^{-md}$. Known Kostant separation therefore survives localization. Local invertibility of θ_m everywhere on this open is unnecessary.

The full normalized Jacobian is

$$\Psi_m(g) = \left(\frac{D(g)}{4^n}\right)^{(nm-1)(2n+1)} \left(1 - nm + \frac{m}{2} \operatorname{tr} g\right).$$

Thus multiplication can cross the additional critical divisor in parentheses. This refines full-Jacobian localization without contradicting the global regularity criterion. It does not remove poles from inverse-differential vector fields. Sharpness of the support $D = 0$, rather than the unit-rescaling observation, is the additional theorem to review.

8 A focused research program

8.1 First: secure the theorem already written

The next deliverable should be a concise external-review packet containing (6), the actual-ideal reduction, the dependency map, and the final E_8 proof. Its questions should be precise:

1. Does the critical presentation preserve the group-fiber scheme, including regular nonsemisimple points and central-cover descent?
2. Do the degree bounds and boundary certificates exclude every required exceptional component, with valid characteristic-zero inference?
3. Is generic smoothness used only after excluding dominating nonreduced components, rather than assuming that exclusion?

The existing package separates algebraic evidence from geometry. A bounded improvement would export ideal-membership multipliers or obtain a second exact check for the engine-dependent step. This improves verifiability, not theorem coverage. No external review has been performed or solicited by this report.

8.2 Second: a classical classification theorem

Target. For arbitrary classical Levi block patterns, classify the components of the full adjoint critical scheme dominating the central target. Express their dimensions, horizontal stabilizers, and constant character values in the block data. Deduce precisely which survive $\chi \mapsto f(\chi)$ for actual polynomial character constructions.

At the time of this review, the unfinished target was the polynomial-adjoint extension: ordinary adjoint nonemptiness and zero excess were already classified in all classical ranks at GEO-49. The all-rank type-A polynomial result and GL_4^q family were anchors for that extension. Success would take a partition or

signed block datum as input and return nonemptiness and excess without separately solving every Levi. The representation-erasure theorem would then become a structural consequence.

1. Write uniform restricted adjoint characters in classical block variables, keeping all linear and constant terms. Establish original-ideal equations in coordinates that retain repeated eigenvalues.
2. Characterize block configurations forcing a constant character on a dominant component; prove the converse or isolate a counterexample to it.
3. Determine stabilizers and dimensions. Invoke the critical-value criterion only after checking actual smoothness.
4. Apply (7), separating surviving constant, erased constant, and nonconstant components. Verify the proposed polynomials really come from representations.

Stopping rule. A failed block criterion must be reported with its exact obstruction. It must not turn into another sequence of unrelated exceptional cases. Exceptional groups should test a mechanism already formulated in general terms.

8.3 Third: sharpen the harmonic question

A focused follow-up would classify the minimal invariant open supporting multiplication by a specified pulled-back harmonic space. Our symplectic family demonstrates that this open can exceed the Jacobian-nonsingular open. A general result needs an orbit-separation obstruction for necessity and an algebraic spanning argument for sufficiency. Existence of some other harmonic complement is a different problem.

This stays close to Kostant–Michor's separation objective. Agricola–Goodman's operators offer a meaningful comparison, but an operator theorem is additional work. We should not call this the solution of an explicitly posed general classification problem unless that formulation is actually located in a source.

8.4 What to defer, and why

Direction	Missing ingredient	Recommendation
Minimum Cayley degree	Lower bounds for all equivariant rational maps	Separate from fixed-map computations.
Uniform exceptional proof	A new full-character identity excluding dominant nonreduced components	Wait for an actual identity.
Numerical Lie methods	Convergence, conditioning, and a demonstrated benefit	Later local-inverse application.
Metaplectic extensions	Phase, composition, and distributional analysis	Require a specific analytic target.
Double-copy applications	A meaningful invariant gain from reparametrization	No immediate project justified.
Graph entropy or torsion	A map-dependent statement linked to our theorem	Do not pursue on citation evidence alone.

9 Novelty and claim boundaries

The reviewed works do not present the same explicit proper-Levi actual-fiber theorem or the proposed block classification. This is weaker than proving that no such result exists anywhere. A submission needs a further targeted comparison with Springer isomorphisms, invariant differential operators, character critical loci, and equivariant morphisms. Absence from the citation corpus is not a priority proof.

Published inputs or elementary consequences: trace projection; degree comparison with the quotient; the invariant-gradient basis; the chain rule; local formal inversion; harmonic separation after invertible homogeneous rescaling on a classical Cayley open. These are not new theorems of our project.

Candidates for our contributions: scheme-level adjoint regular-fiber smoothness by cases; the critical-value stabilizer criterion under its hypotheses; representation-survival and erasure consequences; and sharp invariant support for the fixed symplectic harmonic construction. Each requires its own proof and literature comparison. Computational certificates do not establish novelty.

Not established here: a new minimum Cayley degree; arbitrary-representation generic smoothness; a uniform replacement of the exceptional case proof; the complete polynomial-adjoint block classification; or a result resolving graph, quantization, or double-copy questions. One structural theorem with exact hypotheses and a reviewable proof would be a stronger next advance than another table.

A Two dissertation corrections

The original PDF was visually checked at the pages below. These are corrections to those passages, not a conclusion about every downstream theorem.

A.1 Printed page 27, Theorem 3.1(4): differential discriminant

The passage describes the invertibility locus of $\mathrm{d}\theta_\lambda$ using $\theta_\lambda(h) \neq 0$. The scalar required is

$$\Psi_\lambda(h) = \det(\mathrm{d}\theta_{\lambda,h}),$$

in compatible tangent frames. At the identity of a semisimple group, $\theta_\lambda(e) = 0$ but $\mathrm{d}\theta_{\lambda,e} = 1$, so the printed condition cannot hold. The preceding item introduces Ψ_λ, making a notation substitution the natural correction. Its zero locus is the differential-singular locus, possibly empty.

A.2 Printed page 30, opening paragraph: Jordan decomposition

For $g = g_s g_u$, the semisimple assertion $\theta(g_s) = \theta(g)_s$ is the relevant Kostant–Michor statement. The nilpotent part of $\theta(g)$ is not generally $\theta(g_u)$. The translated map is $u \mapsto \theta(su) - \theta(s)$ and depends on s.

Take defining SL_3, $s = \mathrm{diag}(2, 2, 1/4)$ and $u = I + E_{12}$. They commute and have determinant one. Since $\theta(g) = g - \mathrm{tr}(g)I/3$,

$$\theta(su) - \theta(s) = 2E_{12}, \qquad \theta(u) = E_{12}.$$

The former is the nilpotent part of $\theta(su)$. This disproves the stronger printed assertion and retains the semisimple one. The accompanying rational-matrix script verifies the identities.

B Audit trail

`source-manifest.json` records retrieved full texts, versions, hashes, and the local research notes used

for comparison. Source PDFs remain outside the repository report; the bibliography links to accessible versions. Inclusion does not imply an author's endorsement of our interpretation.

The script `check_comparisons.py` checks the two Lu–Ye examples, the Jordan counterexample, symplectic root values, and the scalar identity on an exact torus. Its results are in `comparison-checks.json`. These checks corroborate the displayed comparisons; they do not replay all existing proofs.

The status snapshot includes uncommitted research notes, identified by individual hashes. Existing proofs and certificate bindings were not rewritten. This report and its source are supplied in an ordinary single-page PDF layout.

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